

AWS GLUE CRAWLER

APARTADO A

1.- Desde Aws CLI explora el contenido del bucket `s3://noaa-ghcn-pds/csv/`.

Utilizando el comando `ls` podemos ver que contiene dos carpetas, y ambas de ellas contienen múltiples archivos `.csv`

```
PS C:\WINDOWS\System32> aws s3 ls s3://noaa-ghcn-pds/csv/
PRE by_station/
PRE by_year/
```

```
PS C:\WINDOWS\System32> aws s3 ls s3://noaa-ghcn-pds/csv/by_station/ |
```

```
2025-02-09 08:54:27      192697 WA010050800.csv
2025-02-09 08:54:27      191686 WA010063530.csv
2025-02-09 08:54:27      158415 WA010066250.csv
2025-02-09 08:54:27      167676 WA010066890.csv
2025-02-09 08:54:27      170768 WA010082900.csv
2025-02-09 08:54:28      177587 WA010085780.csv
2025-02-09 08:54:27      195958 WA010086600.csv
2025-02-09 08:54:27      166587 WA010094670.csv
2025-02-09 08:54:27      192891 WA010095930.csv
2025-02-09 08:54:27      196272 WA010096390.csv
2025-02-09 08:54:27      196151 WA010098500.csv
2025-02-09 08:54:27      190440 WA010098520.csv
2025-02-09 08:54:28      194501 WA010101180.csv
2025-02-09 08:54:29    2498147 WA010101860.csv
2025-02-09 08:54:27      129119 WA010107010.csv
2025-02-09 08:54:28      191378 WA010111300.csv
2025-02-09 08:54:27      129194 WA010117880.csv
2025-02-09 08:54:27      188571 WA010150350.csv
2025-02-09 08:54:27      165087 WA010472480.csv
2025-02-09 08:54:28      482604 WA010517310.csv
2025-02-09 08:54:27      216814 WA010518000.csv
2025-02-09 08:54:27      195641 WA010526410.csv
2025-02-09 08:54:27      196019 WA010542310.csv
2025-02-09 08:54:28      195401 WA010543290.csv
2025-02-09 08:54:27      195356 WA010543500.csv
2025-02-09 08:54:28      858317 WA010553740.csv
2025-02-09 08:54:27      192050 WA010554470.csv
2025-02-09 08:54:27      193355 WA010558090.csv
2025-02-09 08:54:27      197081 WA010574650.csv
2025-02-09 08:54:27      195993 WA010576160.csv
```

```
PS C:\WINDOWS\System32> aws s3 ls s3://noaa-ghcn-pds/csv/by_year/
```

```
PS C:\Users\Mañana> aws s3 ls s3://noaa-ghcn-pds/csv/by_year/
2025-01-14 13:42:27      11608918 1750.csv
2026-01-12 18:29:55       25333 1763.csv
2026-01-12 18:29:56       25340 1764.csv
2026-01-12 18:29:53       25328 1765.csv
2026-01-12 18:29:57       25378 1766.csv
2026-01-12 18:30:01       25374 1767.csv
2026-01-12 18:30:02       25419 1768.csv
2026-01-12 18:30:04       25310 1769.csv
2026-01-12 18:30:04       25316 1770.csv
2026-01-12 18:30:07       25305 1771.csv
2026-01-12 18:30:02       25449 1772.csv
2026-01-12 18:29:54       25305 1773.csv
2026-01-12 18:29:58       25354 1774.csv
2026-01-12 18:30:04      50554 1775.csv
2026-01-12 18:30:04      50705 1776.csv
2026-01-12 18:30:01      50608 1777.csv
2026-01-12 18:30:07      50348 1778.csv
2026-01-12 18:29:57      49631 1779.csv
2026-01-12 18:30:06      50536 1780.csv
2026-01-12 18:30:00      62578 1781.csv
2026-01-12 18:29:54      62494 1782.csv
2026-01-12 18:30:07      62777 1783.csv
2026-01-12 18:30:03      62986 1784.csv
2026-01-12 18:30:01      62786 1785.csv
2026-01-12 18:30:04      62764 1786.csv
2026-01-12 18:30:08      50581 1787.csv
2026-01-12 18:30:06      50633 1788.csv
2026-01-12 18:30:05      62776 1789.csv
```

2.- Descarga uno cualquiera de los archivos que contiene en cada una de sus carpetas y muestra las primeras líneas de ellos.

Utilizamos `cp` para descargar los archivos

```
PS C:\Users\Mañana> aws s3 cp s3://noaa-ghcn-pds/csv/by_station/WA010050800.csv ./
download: s3://noaa-ghcn-pds/csv/by_station/WA010050800.csv to .\WA010050800.csv
```

```
PS C:\Users\Mañana> aws s3 cp s3://noaa-ghcn-pds/csv/by_year/1764.csv ./
download: s3://noaa-ghcn-pds/csv/by_year/1764.csv to .\1764.csv
PS C:\Users\Mañana> |
```

Y utilizamos `Get-Content` con `-First 5` para mostrar las cinco primeras líneas

```

PS C:\Users\Mañana> Get-Content WA010050800.csv -First 5
ID,DATE,ELEMENT,DATA_VALUE,M_FLAG,Q_FLAG,S_FLAG,OBS_TIME
WA010050800,19680101,PRCP,50,,,I,
WA010050800,19680102,PRCP,0,,,I,
WA010050800,19680103,PRCP,0,,,I,
WA010050800,19680104,PRCP,0,,,I,
PS C:\Users\Mañana> Get-Content 1764.csv -First 5
ID,DATE,ELEMENT,DATA_VALUE,M_FLAG,Q_FLAG,S_FLAG,OBS_TIME
ITE00100554,17640101,TMAX,64,,,E,
ITE00100554,17640101,TMIN,44,,,E,
ITE00100554,17640102,TMAX,46,,,E,
ITE00100554,17640102,TMIN,26,,,E,

```

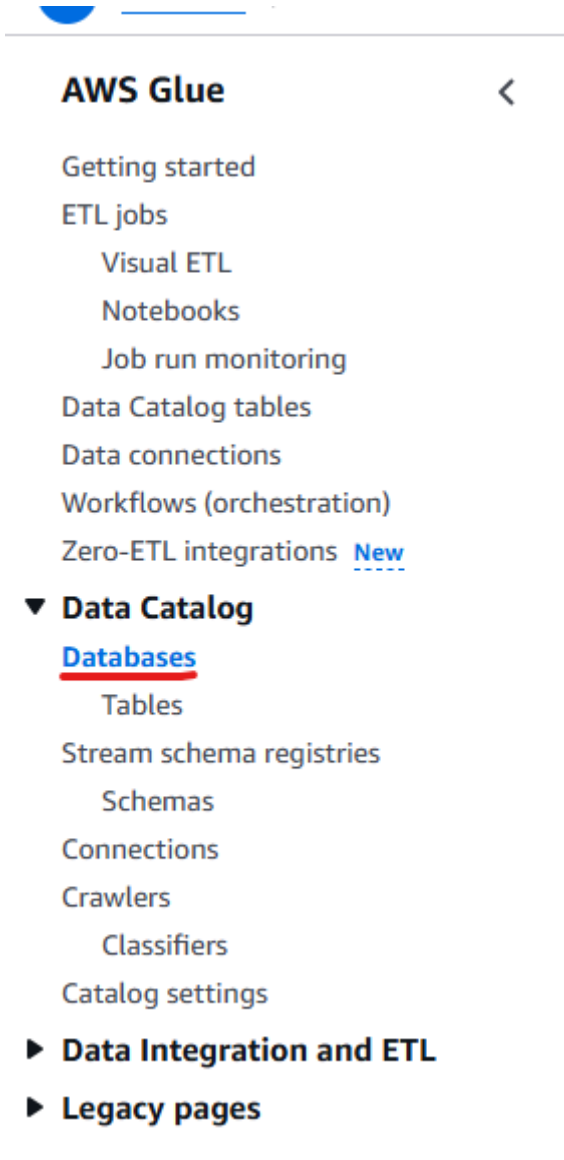
3.- ¿Qué contiene cada uno de los dos tipos de archivos?

Ambos archivos contienen los datos del clima medidos en diferentes estaciones. Los del archivo WA010050800 contiene los de dicha estación, mientras que los del archivo 1764 contiene los de ese año en específico.

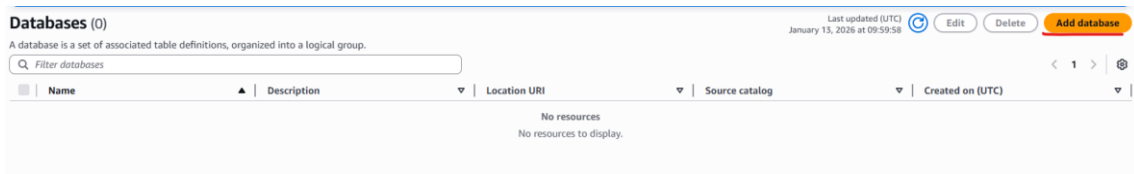
APARTADO B

1.- Crea una base de datos en AWE GLUE llamada clima.

En AWS glue hacemos click en el apartado Data catalog en el menú de la izquierda y después en “Databases”



Después en Add Database



Create a database

create a database in the AWS Glue Data Catalog.

Database details

Name

clima

Database name is required, in lowercase characters, and no longer than 255 characters.

Description - optional

Enter text

Descriptions can be up to 2048 characters long.

Database settings

Location - optional

Set the URI location for use by clients of the Data Catalog.

An S3 location is required for managed tables and Zero-ETL integrations.

Cancel

Create database

Databases (1)

A database is a set of associated table definitions, organized into a logical group.

Q Filter databases

☐	Name	▲	Description
☐	clima		-

2.- Crea un Crawler AWS GLUE que nos explore el bucket del ejercicio anterior generando las tablas en la base de datos que acabas de crear.

De nuevo en el menú de la izquierda en Data Catalog, hacemos click en Crawlers

Zero ETL integrations

▼ Data Catalog

Databases

Tables

Stream schema registries

Schemas

Connections

Crawlers

Classifiers

Catalog settings

► Data Integration and ETL

► Legacy pages

Y después a Create Crawler

Crawlers

A crawler connects to a data store, progresses through a prioritized list of classifiers to determine the schema for your data, and then creates metadata tables in your data catalog.

Crawlers (0)

Info

Last updated (UTC)

January 13, 2026 at 10:07:15

Action

Run

Create crawler

View and manage all available crawlers.

Filter crawlers

Name

State

Schedule

Last run

Last run timestamp

Log

Table changes from last r...

No resources

No resources to display.

Seguimos los pasos que se nos indican

Set crawler properties

Crawler details

Info

Name

clima-gluecrawler

Name can be up to 255 characters long. Some character set including control characters are prohibited.

Description - optional

Enter a description

Descriptions can be up to 2048 characters long.

Tags - optional

Use tags to organize and identify your resources.

Cancel

Next

Choose data sources and classifiers

Data source configuration

Is your data already mapped to Glue tables?

Not yet

Select one or more data sources to be crawled.

Yes

Select existing tables from your Glue Data Catalog.

Data sources (0)

Info

Edit

Remove

Add a data source

The list of data sources to be scanned by the crawler.

Type

Data source

Parameters

You don't have any data sources.

Add a data source

Custom classifiers - optional

A classifier checks whether a given file is in a format the crawler can handle. If it is, the classifier creates a schema in the form of a StructType object that matches that data format.

Cancel

Previous

Next

Add data source



Data source

Choose the source of data to be crawled.

S3



Network connection - optional

Optionally include a Network connection to use with this S3 target. Note that each crawler is limited to one Network connection so any other S3 targets will also use the same connection (or none, if left blank).



Clear selection

Add new connection

Location of S3 data

- ☐ In this account
- ☒ In a different account

S3 path

Browse for or enter an existing S3 path.

s3://noaa-ghcn-pds/csv/

All folders and files contained in the S3 path are crawled. For example, type s3://MyBucket/MyFolder/ to crawl all objects in MyFolder within MyBucket.

Subsequent crawler runs

This field is a global field that affects all S3 data sources.

- ☒ Crawl all sub-folders
Crawl all folders again with every subsequent crawl.
- ☐ Crawl new sub-folders only
Only Amazon S3 folders that were added since the last crawl will be crawled. If the schemas are compatible, new partitions will be added to existing tables.
- ☐ Crawl based on events
Rely on Amazon S3 events to control what folders to crawl.

☐ Sample only a subset of files

☐ Exclude files matching pattern

Cancel

Add an S3 data source

Choose data sources and classifiers

Data source configuration

Is your data already mapped to Glue tables?

☒ Not yet

Select one or more data sources to be crawled.

☐ Yes

Select existing tables from your Glue Data Catalog.

Data sources (1) [Info](#)

The list of data sources to be scanned by the crawler.

[Edit](#)

[Remove](#)

[Add a data source](#)

Type	Data source	Parameters
☒ S3	s3://noaa-ghcn-pds/csv/	Recrawl all

► Custom classifiers - *optional*

A classifier checks whether a given file is in a format the crawler can handle. If it is, the classifier creates a schema in the form of a StructType object that matches that data format.

[Cancel](#)

[Previous](#)

[Next](#)

Configure security settings

IAM role [Info](#)

Existing IAM role

LabRole



[View](#)

[Create new IAM role](#)

[Update chosen IAM role](#)

Only IAM roles created by the AWS Glue console and have the prefix "AWSGlueServiceRole-" can be updated.

Lake Formation configuration - *optional*

Allow the crawler to use Lake Formation credentials for crawling the data source. [Learn more.](#)

☐ Use Lake Formation credentials for crawling S3 data source

Checking this box will allow the crawler to use Lake Formation credentials for crawling the data source. If the data source is registered in another account, you must provide the registered account ID. Otherwise, the crawler will crawl only those data sources associated to the account. Only applicable to S3, Glue Catalog, Iceberg, and Hudi data sources.

► Security configuration - *optional*

Enable at-rest encryption with a security configuration.

[Cancel](#)

[Previous](#)

[Next](#)

Set output and scheduling

Output configuration [Info](#)

Target database

clima



[Clear selection](#)

[Add database](#)

Table name prefix - *optional*

clima_

Maximum table threshold - *optional*

This field sets the maximum number of tables the crawler is allowed to generate. In the event that this number is surpassed, the crawl will fail with an error. If not set, the crawler will automatically generate the number of tables depending on the data schema.

Type a number greater than 0

► Advanced options

Crawler schedule

You can define a time-based schedule for your crawlers and jobs in AWS Glue. The definition of these schedules uses the Unix-like [cron](#) syntax. [Learn more.](#)

Frequency

On demand

[Cancel](#)

[Previous](#)

[Next](#)

Review and create

Step 1: Set crawler properties [Edit](#)

Set crawler properties

Name	Description	Tags
clima-gluecrawler	-	-

Step 2: Choose data sources and classifiers [Edit](#)

Data sources (1) [Info](#)

The list of data sources to be scanned by the crawler.

Type	Data source	Parameters
S3	s3://noaa-ghcn-pds/csv/	Recrawl all

Step 3: Configure security settings [Edit](#)

Configure security settings

IAM role	Security configuration	Lake Formation configuration
LabRole	-	-

Step 4: Set output and scheduling [Edit](#)

Set output and scheduling

Database	Table prefix - optional	Maximum table threshold - optional	Schedule
clima	clima_	-	On demand

[Cancel](#) [Previous](#) [Create crawler](#)

Una vez creado, volvemos a Crawlers, seleccionamos el crawler que acabamos de crear y hacemos click en Run.

Crawlers

A crawler connects to a data store, progresses through a prioritized list of classifiers to determine the schema for your data, and then creates metadata tables in your data catalog.

Crawlers (1/1) [Info](#)

View and manage all available crawlers.

Filter crawlers

☒	Name	State	Schedule	Last run	Last run timestamp	Log	Table changes from last r...
☒	clima-gluecrawler	Ready	-	-	-	-	-

[Action](#) [Run](#) [Create crawler](#)

3.- Desde el apartado de Tablas de AWS GLUE, muestra la descripción del esquema de las tablas detectadas y el resumen de estadístico de sus columnas.

En “Data-catalog”, hacemos click en “Tablas”

- ▼ **Data Catalog**
 - Databases
 - Tables**
 - Stream schema registries
 - Schemas
 - Connections
 - Crawlers
 - Classifiers
 - Catalog settings

► **Data Integration and ETL**

► **Legacy pages**

Elegimos la tabla correspondiente

\table is the metadata definition that represents your data, including its schema. A table can be used as a source or target in a job definition.

Last updated (UTC) Delete Add tables using crawler Add table

[View and manage all available tables.](#)

🔍 [Filter tables](#)

	Name	Database	Location	Classification	Deprecated	View data	Data quality	Column statistics
	clima_csv	clima	s3://noaa-ghcn-pds/csv/	CSV	-	Table data	View data quality	View statistics

clima_csv

Last updated (UTC) January 14, 2026 at 10:08:37 Version 1 (Current version) ▾ Actions ▾

Table overview

Data quality - new

Table details

Name clima_csv	Classification CSV	Deprecated -
Database clima	Location s3://noaa-ghcn-pds/csv/	Column statistics No statistics
Description -	Connection -	
Last updated January 13, 2026 at 11:23:53		

► **Advanced properties**

Schema | Partitions | Indexes | Column statistics - new

Schema (9)

View and manage the table schema.

Q Filter schemas

#	Column name	Data type	Partition key	Comment
1	id	string	-	-
2	date	bigint	-	-
3	element	string	-	-
4	data_value	bigint	-	-
5	m_flag	string	-	-
6	q_flag	string	-	-
7	s_flag	string	-	-
8	obs_time	bigint	-	-
9	partition_0	string	Partition (0)	-

Para ver el resumen estadístico, hacemos click en “Column Statistics”

clima_csv

Table overview

Data quality - new

Table details

Name

clima_csv

Database

clima

Description

-

Last updated

January 13, 2026 at 11:23:53

Classification

CSV

Location

s3://noaa-gl

Connection

-

► Advanced properties

Schema

Partitions

Indexes

Column statistics - new

Después, hacemos click en generate y “generate on demand”

Actions ▼

View all runs

Generate ▲

Generate on schedule

Generate on demand

En las opciones Escogemos “All columns”, le asignamos un rol IAM (En este caso LabRole) y hacemos click en “Generate Statistics”

Column options

All columns

Generate statistics for all columns.

Selected columns

Choose the columns to generate statistics.

IAM Role

Column statistics task requires permissions to read Data Catalog tables and underlying data in the S3 bucket.

IAM role

Create new IAM role

Use an existing IAM role

Existing IAM role

LabRole

View

► Advanced configuration - optional

Cancel

Generate statistics

Una vez terminen de generarse, volvemos a la tabla y observamos las estadísticas en “Column Statistics”

No se han podido enseñar debido a que se bloquea a la hora de realizarlas

4.- ¿Está particionada la tabla? ¿Por qué campos?

En el apartado “Partitions” podemos ver que la tabla está particionada
Por “by_station” y “by_year”

clima_csv

Last updated (UTC)
January 14, 2026 at 10:30:32

Version 1 (Current version)

Actions

Table overview

Data quality - new

Table details

Name
clima_csv

Database
clima

Description
-

Last updated
January 13, 2026 at 11:23:53

Classification
CSV

Location
s3://noaa-ghcn-pds/csv/

Connection
-

Deprecated
-

Column statistics
Stopped

Advanced properties

Schema

Partitions

Indexes

Column statistics - new

Partitions (2)

The list of partitions for this table.

Filter partitions

partition_0

by_station

by_year

Files

View files

View files

Properties

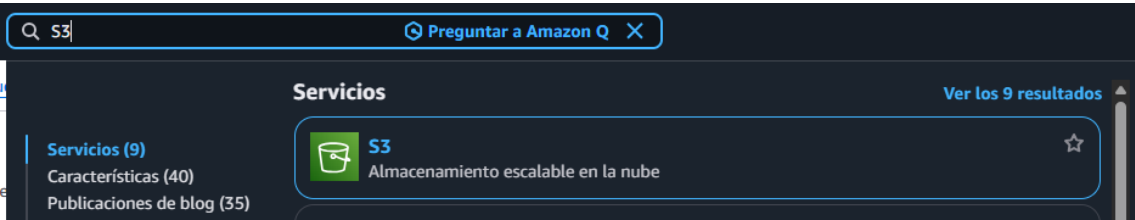
View Properties

View Properties

APARTADO C

Desde ATHENA, intenta realizar las siguientes consultas mostrando sus resultados y tiempos de ejecución:

Primero configuramos el bucket para guardar las consultas



Configuración general

Región de AWS
EE.UU. Este (Norte de Virginia) us-east-1

Tipo de bucket Información

☒ **Uso general**
Recomendado para la mayoría de los casos de uso y patrones de acceso. Los buckets de uso general son del tipo de bucket de S3 original. Permiten una combinación de clases de almacenamiento que almacenan objetos de forma redundante en múltiples zonas de disponibilidad.

☐ **Directorio**
Recomendado para casos de uso de baja latencia. Estos buckets utilizan únicamente la clase de almacenamiento S3 Express One Zone, que proporciona un procesamiento más rápido de los datos dentro de una única zona de disponibilidad.

Nombre del bucket Información

consultasclima

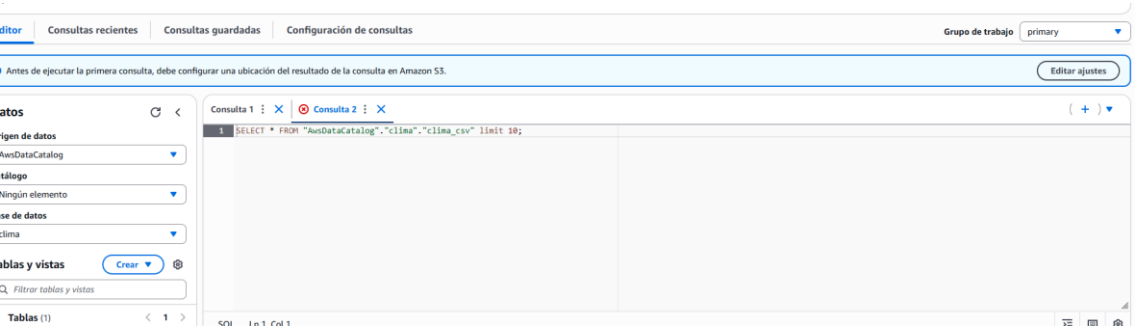
Los nombres de los buckets deben tener entre 3 y 63 caracteres y ser únicos dentro del espacio de nombres global. Los nombres de los buckets también deben empezar y terminar con una letra o un número. Los caracteres válidos son a-z, 0-9, puntos (.) y guiones (-). [Más información](#)

Copiar la configuración del bucket existente: *opcional*
Solo se copia la configuración del bucket en los siguientes ajustes.

[Elegir el bucket](#)

Formato: s3://bucket/prefijo

Después entramos a Athena desde “Table Data”



Para asignar donde se guardarán las consultas, entramos en Configuración de consultas y después en Administrar

Editor

Consultas recientes

Consultas guardadas

Configuración de consultas

Cifrado de resultados de consultas	Administrar
Ubicación del resultado de la consulta	Cifrado de los resultados de la consulta
-	-
Propietario previsto del bucket	Asigna al propietario del bucket el control total sobre los resultados de la consulta
-	Desactivado

Ahora asignamos el bucket que hemos creado

Administre la ubicación y codificación de los resultados de la consulta



Location of query result - *optional*

Enter an S3 prefix in the current region where the query result will be saved as an object.

s3://bucket/prefix/object/

View

[Browse S3](#)

Expected bucket owner - *optional*

Specify the AWS account ID that you expect to be the owner of your query results output location bucket.

Enter AWS account ID

☐ **Assign bucket owner full control over query results**

Enabling this option grants the owner of the S3 query results bucket full control over the query results. This means that if your query result location is owned by another account, you grant full control over your query results to the other account.

☐ **Encrypt query results**

Cancelar

Guardar

Choose S3 data set



S3 buckets

Bucket (1/4)



Filter bucket

< 1 >

Name	Creation date
☐ aws-logs-777558146599-us-east-1	2026-01-12T09:08:48.000+01:00
☒ consultascima	2026-01-14T12:18:55.000+01:00
☐ demo-bucket-pff	2025-12-10T11:09:20.000+01:00
☐ hdf-bucket	2025-12-18T10:40:55.000+01:00

Cancel

Choose

Administre la ubicación y codificación de los resultados de la consulta



Location of query result - *optional*

Enter an S3 prefix in the current region where the query result will be saved as an object.

🔍 s3://consultasclima



View [↗](#)

Browse S3



You can create and manage lifecycle rules for this bucket

Use Amazon S3 lifecycle rules to store your query results and metadata cost effectively or to delete them after a period of time. [Learn more](#) [↗](#)

Lifecycle configuration [↗](#)

Expected bucket owner - *optional*

Specify the AWS account ID that you expect to be the owner of your query results output location bucket.

Enter AWS account ID



Assign bucket owner full control over query results

Enabling this option grants the owner of the S3 query results bucket full control over the query results. This means that if your query result location is owned by another account, you grant full control over your query results to the other account.



Encrypt query results

Cancelar

Guardar

1.- ¿Cuántos registros tiene la tabla?

Consulta 1 [×](#)

✓ Consulta 2 [×](#)

```
1 SELECT COUNT(*) FROM "AwsDataCatalog"."clima"."clima_csv";
```

Tiempo en cola: 104 ms Tiempo de ejecución: 12.092 sec Datos analizados: 206.68 GB

Resultados de la consulta

Estado de la consulta

✓ Completado

Resultados (1)

🔍 Filas de búsqueda

▾ | _col0

#	_col0
1	6336799722

2.- ¿Cuántas mediciones tenemos de España?

Consulta 1 : ✕ | ✓ Consulta 2 : ✕ | ✓ Consulta 3 : ✕

```
1 SELECT COUNT(*) FROM "clima"."clima_csv" where id like 'SP%';
```

Tiempo en cola: 2.863 sec Tiempo de ejecución: 16.501 sec Datos analizados: 206.68 GB

Resultados de la consulta

Estado de la consulta

✓ Completado

Resultados (1)

🔍 Filas de búsqueda

▾ | _col0

#	_col0
1	21166382

3.- Sabiendo los códigos de las 4 estaciones de Asturias ¿Cuántas mediciones tenemos de Asturias?

Consulta 1 : ✕ | Consulta 2 : ✕ | Consulta 3 : ✕ | Consulta 4 :

1 SELECT COUNT(*) FROM "clima"."clima_csv" where id like 'SPE00119792'

2 OR id like 'SPE00119801'

3 OR id like 'SPE00119819'

4 OR id like 'SPE00119828';

Tiempo en cola: 109 ms Tiempo de ejecución: 12.659 sec Datos analizados: 206.68 GB

Resultados de la consulta | Estado de la consulta

Completado

Resultados (1)

Q Filas de búsqueda

▾ | _col0

1 544046

4.- ¿Cuántas mediciones tenemos de Oviedo?

Consulta 5 :

1 SELECT COUNT(*) FROM "clima"."clima_csv" where id like 'SPE00119828';

Tiempo en cola: 132 ms Tiempo de ejecución: 30.021 sec Datos analizados: 206.68 GB

▾ | _col0

1 146094

5.- ¿Cuál es la medición más antigua de España, Asturias y Oviedo?

Consulta 5 : ✕ | **Consulta 6 : ✕** | Consulta 11 : ✕

```
1
2 (SELECT * FROM "clima"."clima_csv" WHERE id like 'SP%' ORDER BY date LIMIT 1)
3 UNION
4 (SELECT * from "clima"."clima_csv" WHERE id like 'SPE00119792'
5 OR id like 'SPE00119801'
6 OR id like 'SPE00119819'
7 OR id like 'SPE00119828' ORDER BY date LIMIT 1)
8 UNION ALL
9 (SELECT * FROM "clima"."clima_csv" WHERE id like 'SPE00119828' ORDER BY date LIMIT 1)
```

Tiempo en cola: 109 ms | Tiempo de ejecución: 43.033 sec | Datos analizados: 620.04 GB

#	id	date	element	data_value	m_flag	q_flag	s_flag	obs_time	partition_0
1	SPE00119828	19721201	TMAX	130			E		by_year
2	SPE00119801	19381001	TMAX	192			E		by_year
3	SPE00155329	18961101	TMAX	155			E		by_year